Lecture 12: Evolution, Emotion, and Reason: Emotions, Part II

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-12 (CC BY-NC-SA 3.0).

Before you start

This lecture continues the study of emotions as evolutionary adaptations. Professor Bloom finishes the topic of love between parents and children, then asks a harder question: why are animals and people kind to those who are not their relatives? He uses two famous games to answer it and ends with a cultural difference: cultures of honor.

Learning goals

After this lecture you should be able to:

- Describe Harlow's wire mother and cloth mother studies and explain what they show about attachment.
- Explain reciprocal altruism and why cheaters (free-riders) are a problem for it.
- Describe the structure of the Prisoner's Dilemma and explain why Tit-for-Tat worked so well.
- Explain why being irrational can sometimes be an advantage, using the Ultimatum Game.
- Describe what a culture of honor is and what the University of Michigan study found.

Key ideas

1. Emotional Responses to Caregivers and Kin

- As far as Bloom knows, nonhuman primate smiles are almost all appeasement ("don't hurt me") smiles, with no greeting smile and no Duchenne smile.
- Humans invest in quality, not quantity, so each child is precious. Bloom looks at parental love in another species: Emperor penguins in the film "March of the Penguins." A Darwinian sees their complex care for eggs as a biological adaptation, because children share their parents' genes.
- Babies become attached to whoever is closest, usually the mother. Bloom calls the idea of one magic moment of "imprinting" after birth "silly."
- The "Cupboard Theory" says babies love the mother because she gives food. Bowlby says babies want comfort and social interaction. Harlow's baby monkeys ate from the wire mother but loved the cloth mother, and used it as a base when threatened and when exploring.
- On daycare, nobody really knows the subtle effects, but it does not make a big difference.
- No contact at all is different. Monkeys raised alone with only a wire mother became withdrawn, did not play, bit themselves, and were incompetent socially, sexually and as mothers. Children in harsh

orphanages develop severe social and emotional problems. Early help can reverse this: a younger "monkey therapist" helped isolated monkeys improve.

2. Question and Answer on Emotions towards Kin

A teaching assistant said there is evidence that support among the young can help both monkeys and children. For the "middle ground" (parents who are not very affectionate), there is no good evidence of big, dramatic effects.

3. Evolutionary Explanations for Emotional Responses

- Love for kin is not a big evolutionary puzzle. Kindness to non-kin is. Animals groom each other, give warning cries, and share food. Vampire bats that find a lot of blood vomit it into the mouths of other bats.
- This is "reciprocal altruism": I am kind to you because I expect you to be kind to me.
- The problem is cheaters, or "free-riders," who take the benefits without paying the costs. A bat that accepts blood but never shares would do better than a bat that shares. At a bar, it pays to accept drinks but never buy a round.
- The solution is "cheater detection": animals must recognize, remember and punish cheaters. In an experiment, scientists stopped one vampire bat from sharing. Later, the other bats refused to feed the "selfish" bat.

4. Cooperative Behavior and the Prisoner's Dilemma

- Two friends rob a bank and are caught. Each can cooperate (stay silent) or defect (squeal). The best case for you is to defect while the other cooperates; the worst is the opposite.
- Whatever your friend does, you are better off defecting. But if both defect, both are worse off than if both cooperate. That is the tragedy of the game. The same structure appears in a divorce, in nuclear weapons, and in a drug deal.
- Robert Axelrod held a computer competition with sixty-three programs. The winner, by Anatol Rapoport, was one of the simplest: "Tit-for-Tat." It cooperates the first time, then copies what the other program did on the previous trial. It is friendly, not a sucker, forgiving, and transparent (easy to understand).
- Some psychologists argue that emotions match the game: we like people who cooperate with us, we feel anger and distrust towards those who betray us, and we feel bad when we betray someone who cooperated.

5. The Ultimatum Game, Rationality and Irrationality

- Player "A" has ten dollars and offers "B" any amount from one to ten. If "B" accepts, both keep their share. If "B" rejects, nobody gets anything.
- A rational "B" should accept one dollar, because one dollar is better than nothing. But real people reject unfair offers "just out of spite," so you need to offer more.
- A rational person is easily exploited. A person with a temper, or a reputation for mild irrationality, gets treated better. The same idea may explain love: someone "head over heels in love" is irrational, but you can trust them more.

- Daly and Wilson report that most murder comes from insults, curses and small offenses. Bloom calls this "adaptive irrationality."

6. Cultures of Honor

- A culture of honor appears when you cannot rely on the law and your resources, such as cattle, are easily taken. Then a reputation for violent retaliation matters. Examples: Scottish highlanders, Masai warriors, Bedouin tradesmen, and Western cowboys.
- The most studied example is the American South. Richard Nisbett and Dov Cohen found more approval of corporal and capital punishment and more permissive gun laws there. Violence was higher only for crimes of honor.
- In a study with University of Michigan students, a graduate student bumped into white male participants in a hallway and insulted them. On average, southern men showed a bigger hormone and stress response (increases in testosterone and cortisol) than northern men.
- Bloom adds that these are average differences, and honor cultures are not simply worse: Nisbett, a southerner, was surprised by how rude northerners were.
- Conclusion: emotions are not noise but complex motivational systems, shaped by evolution and modified by culture.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
attachment	a strong emotional bond to a caregiver	Baby monkeys loved the cloth mother
imprinting	the idea of one magic moment when a baby bonds	Bloom calls this idea "silly"
non-kin	people or animals you are not related to	Animals give warning cries, often to protect non-kin
reciprocal altruism	being kind because you expect kindness back	Vampire bats sharing blood
free-rider (cheater)	someone who takes benefits without paying costs	Accepting drinks but never buying a round
cheater detection	recognizing, remembering and punishing cheaters	The bats starve the "selfish" bat
cooperate / defect	work with the other person / betray them	Staying silent / squealing to the police
Prisoner's Dilemma	a game where each person gains by defecting, but both lose if both defect	Two countries both building weapons
Tit-for-Tat	cooperate first, then copy the other's last move	Anatol Rapoport's winning program
Ultimatum Game	one player offers a split; the other accepts or rejects it	"A" offers "B" one dollar out of ten

Term	Meaning in simple English	Example from the lecture
rational / irrational	following / not following careful logic	Rejecting one dollar "just out of spite"
culture of honor	a culture where a reputation for violent revenge matters	The American South

Check your understanding

1. What did Harlow's baby monkeys do with the wire mother and the cloth mother? Which theory does this support?
2. What happened to monkeys raised with no social contact?
3. Why is kindness to non-kin a bigger evolutionary puzzle than love for our children?
4. What is a free-rider? How did vampire bats respond to one in the experiment?
5. In the Prisoner's Dilemma, why is it always better for you to defect? Why is this "tragic"?
6. Name four features that made Tit-for-Tat successful.
7. You are "B" in the Ultimatum Game, and "A" offers you one dollar out of ten. What would a purely rational person do, and what do real people often do?
8. What two conditions lead to a culture of honor? What did the Michigan hallway study find?

Answer guide

1. They ate from the wire mother but loved the cloth mother and used it as a safe base. This supports Bowlby (comfort), not the Cupboard Theory (food).
2. They became withdrawn, did not play, bit themselves, and were incompetent socially, sexually and as mothers.
3. Kin share our genes, so caring for kin helps copies of our genes. Helping non-kin does not, and cheaters can take advantage of it.
4. A free-rider takes benefits without paying costs. When a bat was stopped from sharing, the other bats later refused to feed it.
5. Whatever the other person does, defecting gives you a better result. But if both defect, both are worse off than if both had cooperated.
6. It starts friendly, it is not a sucker, it is forgiving, and it is transparent.
7. A purely rational person accepts, because one dollar is better than nothing. Real people often reject unfair offers out of spite.
8. You cannot rely on the law, and resources are easily taken. After being insulted, southern men showed a bigger hormone and stress response than northern men.

English practice

Useful phrases from the lecture

- "The short answer is, nobody really knows." - gives a quick, honest answer.
- "Here is the important structure of..." - introduces the key logic of an idea.
- "Once you start thinking about things this way, there's no end to..." - says an idea fits many situations.
- "This is faintly paradoxical, but..." - introduces an idea that seems contradictory but is true.
- "In any case, ..." - moves to the main conclusion after side points.

Speaking task (with your AI tutor)

- Explain the Prisoner's Dilemma in 4-6 sentences. Use: *cooperate*, *defect*, *Tit-for-Tat*.

Writing task

- Write one paragraph (80-120 words): How do vampire bats show both reciprocal altruism and cheater detection? Describe what the bats do and what the experiment found.

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 12. Ask me questions about reciprocal altruism and cheaters, one at a time."
- "Quiz me on the key vocabulary of Lecture 12."

- "Play the Prisoner's Dilemma with me for five rounds, then ask me to explain my strategy."
- "I will explain cultures of honor in my own words. Check my understanding and my English."

Readings from the original course

- Gray, Peter. Psychology (5th edition), pp. 68-75, 212-223, 522-527
- Ekman, Paul. "Emotions Revealed: Recognizing Faces and Feelings to Improve Communication and Emotional Life." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 245-254

My notes